TRADEFLOW · PHASE 3D — REGIME-CONDITIONAL SIZING

Manage the Downside Instead of Fixing the Exit

Final lever in the Phase 3 sequence — accept choppy markets will underperform, reduce exposure during them rather than trying to make them profitable

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> Score floor (Phase 3) and exit logic (Phase 3B/3C) both tried to improve what happens *within* choppy markets and both moved Window 5 only slightly without beating the fixed-exit combined baseline (+23.6%). This lever takes a structurally different approach: scale position size down (continuously, by regime tier) rather than trying to make choppy trades individually more profitable. Three tiers — strong trend 1.0x, weak trend 0.75x, ranging/low-vol 0.5x — applied as a risk-percentage multiplier at the same point the regime classifier already runs for every signal.

Architecture Check

> `effectiveRiskPct` is already computed **per-signal**, inside the main simulation loop, starting from `config.riskPerTrade` and then adjusted by the existing capital-protection tiers — it is not a single value read once per run. `regime` is classified earlier in the same loop iteration and was already in scope at the position-sizing point. The minimal implementation: one new optional field, `regimeSizeMultipliers?: Partial<Record<MarketRegime, number>>`, applied as `effectiveRiskPct *= regimeSizeMultipliers?.[regime] ?? 1.0` immediately after the existing capital-protection tier logic — composing multiplicatively with it rather than replacing it. Undefined/missing keeps today's behavior unchanged for every existing caller. A small JSON-override textarea was added to `/simulate` (mirroring the one already on `/auto-sim`) since this is a nested config object without a natural single-control UI element.

> Verified directly before running the batch: a smoke test against a real window showed average capital-at-risk by classified regime of `strong_uptrend $1582 → weak_uptrend $1079 → weak_downtrend $933 → ranging $584` — confirming the multiplier is actually being applied and scales down through the tiers as designed, not just configured but inert.

Two-Condition Comparison

CONDITION	MEAN RETURN	MEAN PF	MEAN WR	PROFITABLE SUB-RUNS	MEAN MAX DD	MEAN TRADES
A — floor 60, fixed exit, 1x size (baseline)	+20.4%	1.12	51.0%	10/18 (56%)	19.5%	173.4
E — regime-conditional sizing	+15.0%	1.13	50.7%	**13/18 (72%)**	**14.6%**	183.6

On 15m alone: lower average return (-5.4pp), essentially identical PF and win rate, but a **materially higher profitable-run rate (72% vs 56%)** and a **meaningfully lower mean drawdown (14.6% vs 19.5%, a 25% relative reduction)**. This is a real risk/return trade-off, not a clean win or a clean loss — exactly the kind of result this lever was designed to produce.

Per-Window Return and Drawdown

WINDOW	REGIME	A RETURN	E RETURN	A MAX DD	E MAX DD	E MEAN SIZE
1 (2020-10→2021-03)	Bull	+21.5%	+19.2%	13.7%	11.1%	0.75x
2 (2023-10→2024-03)	Bull	+27.3%	+17.9%	21.3%	15.1%	0.69x
3 (2021-05→2021-10)	Bear	+22.1%	+11.4%	22.4%	20.0%	0.74x
4 (2022-04→2022-09)	Bear	+31.3%	+30.3%	18.7%	12.6%	0.71x
5 (2023-07→2023-12)	**Choppy**	-14.9%	**-4.6%**	**23.8%**	**15.4%**	**0.65x**
6 (2024-04→2024-09)	Choppy (sub.)	+35.2%	+16.1%	17.5%	13.6%	0.69x

Window 5 Isolated — The Strongest Single Result in All of Phase 3

-14.9% → -4.6%	23.8% → 15.4%	0.65x	Lowest of 6
WINDOW 5 RETURN (A → E)	WINDOW 5 MAX DD (A → E)	WINDOW 5 MEAN SIZE	VS. ALL OTHER WINDOWS

Regime-conditional sizing moved Window 5's return by +10.3 percentage points and its drawdown by -8.4 points — far larger than anything score floor (+1.4pp on return, did not move floor 65 vs 60) or exit logic (+2.2 to +2.9pp) achieved. Window 5 also has the lowest mean size multiplier of all six windows (0.65x), confirming the regime classifier correctly detected this as the most ranging/weak-trend-heavy window in the set — the lever had real exposure to act on here, unlike a misclassification scenario where it would have nothing to grip.

Proportional Movement

WINDOW 5 Δ VS A (PP)	WINDOWS 1-4+6 AVG Δ VS A (PP)
+10.3	-8.5

The strongest proportional divergence seen in the entire Phase 3 sequence — Window 5 improved while every other window gave up return, in both directions far exceeding what either score floor or exit logic produced. This is the clearest confirmation yet that the lever is targeting the right mechanism.

Regime Breakdown — Where Did Size Reductions Actually Fire?

WINDOW	MEAN SIZE MULTIPLIER	NOTE
1 (Bull)	0.746	Highest of the 6 — most "strong trend" exposure
2 (Bull)	0.687	
3 (Bear — classifier-disagreement window)	0.741	**See caveat below**
4 (Bear)	0.710	
5 (Choppy)	**0.653**	**Lowest — most size reduction, correctly targeted**
6 (Choppy sub.)	0.687	

Window 3 is the one place this lever's blind spot shows up exactly where predicted. The Baseline Campaign flagged that `classifyRegime()` labeled this known China-mining-ban-crash bear window as "weak_uptrend" rather than a downtrend — and that mislabeling carries through here: Window 3's mean size multiplier (0.741) is nearly as high as Window 1's cleanest bull trend (0.746), meaning **the sizing lever barely fired during a window that, by hand-picked label, should have warranted real caution.** This is not a new problem introduced by this batch — it's the same classifier imprecision the Baseline Campaign already identified, now showing up as a concrete cost: Window 3's return dropped substantially under E (22.1%→11.4%) without getting a meaningful drawdown benefit in exchange (22.4%→20.0%, a much smaller improvement than every other window saw).

A broader nuance worth flagging: no window's mean size multiplier rises above 0.75x, even the cleanest bull trend (Window 1). The "three discrete tiers" design (1.0x / 0.75x / 0.5x) intends sharp differentiation, but in practice `strong_uptrend` / `strong_downtrend` classifications are relatively rare against `weak_*` and `ranging` — most signals fire during weak-trend or ranging conditions regardless of the window's overall hand-picked character. The realized effect is closer to "a fairly uniform ~25-35% size haircut almost everywhere, deeper in the choppiest window" than "full size in trends, half size in chop." It is still directionally correct and still concentrated most heavily on Window 5 — just less sharply differentiated across windows than the discrete tier labels might suggest.

Ceiling Check — A Real, Substantial Trade-Off

CONDITION	BEST INDIVIDUAL SUB-RUN
A	+108.0% (SOLUSDT, Window 6)
E	+62.9% (SOLUSDT, Window 2)

Unlike the exit-logic variants (which preserved the ceiling almost exactly), regime-conditional sizing cuts the ceiling substantially — the same SOLUSDT/Window 6 run that produced A's best result (+108.0%) only returned +16.1% on average under E, and a different run (SOLUSDT, Window 2) became the new best at +62.9%. The mechanism: a big winning run is typically a long sequence of trades riding a sustained move, and that sequence inevitably includes stretches the classifier reads as weak-trend or ranging even within an overall strong move — sizing those down compounds across the whole sequence. **This is the real cost of this lever: it doesn't just trim losses, it trims the best wins too.**

Drawdown Comparison

CONDITION	MEAN MAX DD	WORST SINGLE DD	WINDOW 5 MAX DD	WINDOW 5 WORST SUB-RUN DD
A	19.5%	29.5%	23.8%	24.7%
E	**14.6%**	**26.7%**	**15.4%**	**18.6%**

> This is the strongest argument for adopting this lever. Every single drawdown metric improved, and Window 5's drawdown improvement (-8.4 points) is the largest single-window risk reduction seen anywhere in Phase 3.

Sharpe Ratio

CONDITION	MEAN SHARPE (PER SUB-RUN, UNANNUALIZED)
A	0.56
E	**0.65**

E's Sharpe is higher despite lower raw return — the volatility/drawdown reduction outweighs the return reduction in risk-adjusted terms. The standard deviation of the 6 windows' combined (15m+1h) returns also drops from 15.9 to 12.5 under E, confirming this is a genuine consistency improvement, not just a lower number presented favorably.

Combined 15m + 1h Projected Impact

CONFIGURATION	COMBINED MEAN RETURN	WINDOWS ≥30%	COMBINED MEAN MAX DD
A — Floor 60 + 1h (current best)	+23.6%	2/6	15.5%
E — Regime sizing + 1h	+20.9%	2/6	**13.1%**

> Combined return drops by 2.7 points (+23.6%→+20.9%) while combined drawdown improves by 2.5 points (15.5%→13.1%). Neither configuration changes how many windows clear the 30% bar (2/6 either way).

Explicit Verdict

> **OUTCOME 2 applies: regime-conditional sizing reduced return (from +23.6% to +20.9% combined, a real -2.7pp give-up, not a trivial one) but materially improved drawdown (from 15.5% to 13.1% combined, and from 23.8% to 15.4% in Window 5 specifically — the largest single-window risk reduction in the whole Phase 3 sequence). Sharpe improved (0.56 → 0.65).** The trade-off is genuine in both directions: this lever does the best job yet of protecting Window 5, but it also cuts the ceiling on the best winning runs substantially (+108.0%→+62.9% best case) and barely fires where the regime classifier mislabels a window (Window 3, the same classifier blind spot the Baseline Campaign already flagged).

> **Recommendation: adopt Condition E if drawdown reduction and consistency are valued over raw return; keep Condition A (fixed-exit floor-60, no sizing scale) if raw return is the only metric that matters.** Either way, **Phase 3 levers are now exhausted** — three approaches tested (entry quality via score floor, exit mechanics via trailing stops, and now exposure management via regime-conditional sizing) and none combine to clear 30% outright, though each has incrementally improved some dimension of the strategy. **Proceed to fresh-window validation next** (2-3 new 6-month windows never used in any Phase 3 batch), run against whichever of A or E is chosen as the carried-forward config, to confirm these improvements generalize before the go/no-go decision on live trading infrastructure.

Appendix — All 36 Sub-Run Results (Conditions A, E)

WINDOW	SYMBOL	CONDITION	TRADES	WR	PF	RETURN	MAX DD	MEAN SIZE MULT.
1	BTCUSDT	A	164	52.4%	1.15	+18.5%	16.6%	n/a*
1	ETHUSDT	A	177	55.4%	1.07	+8.0%	12.2%	n/a*
1	SOLUSDT	A	137	56.2%	1.31	+38.0%	12.3%	n/a*
1	BTCUSDT	E	170	51.8%	1.18	+17.4%	14.3%	0.74x
1	ETHUSDT	E	181	55.2%	1.02	+2.2%	10.0%	0.72x
1	SOLUSDT	E	144	56.9%	1.41	+37.9%	9.0%	0.78x
2	BTCUSDT	A	191	46.6%	0.77	-23.7%	24.6%	n/a*
2	ETHUSDT	A	171	49.1%	0.98	-2.0%	23.6%	n/a*
2	SOLUSDT	A	186	57.0%	1.68	+107.6%	15.6%	n/a*

2	BTCUSDT	E	204	46.6%	0.81	-15.9%	18.0%	0.64x
2	ETHUSDT	E	184	50.5%	1.07	+6.5%	14.5%	0.69x
2	SOLUSDT	E	198	56.1%	1.55	+62.9%	12.8%	0.73x
3	BTCUSDT	A	181	47.0%	0.82	-20.7%	28.4%	n/a*
3	ETHUSDT	A	153	56.2%	1.34	+50.8%	24.6%	n/a*
3	SOLUSDT	A	149	61.7%	1.33	+36.2%	14.1%	n/a*
3	BTCUSDT	E	190	45.8%	0.78	-21.1%	26.7%	0.72x
3	ETHUSDT	E	154	55.2%	1.34	+34.3%	22.0%	0.76x
3	SOLUSDT	E	159	61.6%	1.24	+20.8%	11.2%	0.75x
4	BTCUSDT	A	177	45.8%	0.95	-6.7%	23.6%	n/a*
4	ETHUSDT	A	186	54.8%	1.39	+55.6%	17.5%	n/a*
4	SOLUSDT	A	201	53.2%	1.23	+45.1%	14.9%	n/a*
4	BTCUSDT	E	193	44.6%	1.00	+0.2%	16.2%	0.68x
4	ETHUSDT	E	193	54.4%	1.38	+38.3%	13.5%	0.72x
4	SOLUSDT	E	211	53.6%	1.37	+52.3%	8.1%	0.74x
5	BTCUSDT	A	172	43.0%	0.93	-9.1%	23.3%	n/a*
5	ETHUSDT	A	150	41.3%	0.79	-21.5%	24.7%	n/a*
5	SOLUSDT	A	188	44.1%	0.89	-14.2%	23.3%	n/a*
5	BTCUSDT	E	184	42.4%	0.93	-6.3%	18.6%	0.62x
5	ETHUSDT	E	155	42.6%	0.87	-9.2%	13.6%	0.64x
5	SOLUSDT	E	202	44.6%	1.02	+1.7%	14.0%	0.70x
6	BTCUSDT	A	174	54.6%	1.21	+26.7%	12.9%	n/a*
6	ETHUSDT	A	170	41.8%	0.69	-29.0%	29.5%	n/a*
6	SOLUSDT	A	194	58.0%	1.66	+108.0%	10.0%	n/a*
6	BTCUSDT	E	187	52.4%	1.11	+8.9%	12.6%	0.66x
6	ETHUSDT	E	181	43.6%	0.79	-18.7%	19.0%	0.68x
6	SOLUSDT	E	214	55.4%	1.48	+58.0%	9.3%	0.71x

* Condition A used 1.0x sizing throughout (no regime-conditional scale), so a per-trade "mean size multiplier" wasn't applicable/captured for it.

Report generated from 18 newly-run live simulations (Condition E) plus existing saved data for Condition A (Phase 3 Score Floor Test), against real historical Binance market data. Zero console errors. Raw data: `training-exports/phase3d-conditionE-results.json` , `training-exports/phase3-floor60-results.json` .